Carlos Payá — Curriculum Vitae

Email	carlos.paya@csic.es	**Affiliation**	ICMM-CSIC, Madrid, Spain
Website	carlosp24.github.io	**Nationality**	Spanish
ORCID	0000-0001-5709-2290	**Languages**	Spanish (native), English & French (C1)
Github	CarlosP24	**Latest update**	May 19, 2026

Personal Profile

Theoretical condensed matter physicist completing my PhD (expected late 2026/early 2027) at the QUDYMA and Q4Q groups at ICMM-CSIC, Madrid, under Elsa Prada and Ramón Aguado.

My research focuses on one-dimensional hybrid semiconductor-superconductor systems, with particular emphasis on full-shell nanowires, combining large-scale numerical simulations with analytical modelling to address open questions spanning topological superconductivity, transport phenomena, and the rich phase structure of these devices.

I have published eight papers in peer-reviewed journals, with first or co-first authorship on six of them. My work has been carried out in close collaboration with leading experimental groups, including those of C. M. Marcus, S. Vaitieknas, Eduardo Lee, and Jesper Nygård. I also undertook a research visit to Karsten Flensberg's group at the Niels Bohr Institute (April–July 2025), where I initiated a new research direction on non-Hermitian transport in hybrid devices.

Education

2023-Present — **PhD in Condensed Matter Physics** - Universidad Autónoma de Madrid, Madrid, Spain

Tentative completion date	Late 2026 / Early 2027
Tentative title	*Full-shell Hybrid Nanowires*
Supervisors	Elsa Prada and Ramón Aguado

2021-2022 — **Master in Condensed Matter Physics** - Universidad Autónoma de Madrid, Madrid, Spain

Grade	9.45/10
Master's Thesis	*Topological phase and Majorana zero modes in full-shell nanowires*
Supervisor	Elsa Prada

2017-2021 — **Bachelor in Physics** - Universidad Autónoma de Madrid, Madrid, Spain

Grade	8.44/10
Bachelor's Thesis	*Josephson junctions based on full-shell Majorana nanowires*
Supervisor	Elsa Prada

Employment History

March 2023 - Present — Instituto de Ciencia de Materiales de Madrid (ICMM), CSIC, Madrid, Spain
PhD Candidate, FPI grant

Quantum Dynamics of Materials (QUDYMA)
and Quantum Materials for Quantum Technologies (Q4Q) groups.

- **April - July 2025** — Niels Bohr Institute, University of Copenhagen, Denmark
  Visiting PhD candidate

  Karsten Flensberg's group.

Nov 2021 - Sep 2022 — Instituto de Ciencia de Materiales de Madrid (ICMM), CSIC, Madrid, Spain
Research Assistant

Theory of Quantum Materials and Solid State Quantum Technologies group.

Publications

Preprints

[P9] Absence of Quasi-Majorana False Positives in Full-Shell Hybrid Nanowires
C. Payá, C. Robles, P. San-Jose, and E. Prada

Journals

* indicates first theory author.

[P8] Non-Hermitian skin effect and electronic nonlocal transport
C. Payá, O. Solow, E. Prada, R. Aguado, and K. Flensberg
Physical Review B, 113, L161405 (2026), arXiv:2510.00921
Citations: 1

[P7] Fluxoid valve effect in full-shell nanowire Josephson junctions
C. Payá, F. J. Matute-Cañadas, A. L. Yeyati, R. Aguado, P. San-Jose, and E. Prada
Physical Review B, 112, 134520 (2025), arXiv:2504.16989
Citations: 3

[P6] Josephson effect and critical currents in trivial and topological full-shell hybrid nanowires
C. Payá, R. Aguado, P. San-Jose, and E. Prada
Physical Review B, 111, 235420 (2025), arXiv:2503.09756
Citations: 5

[P5] Caroli–de Gennes–Matricon Analogs in Full-Shell Hybrid Nanowires
M. T. Deng, **C. Payá***, P. San-Jose, E. Prada, C. M. Marcus, and S. Vaitieknas
Physical Review Letters, 134, 206302 (2025), arXiv:2501.05419
Citations: 7

[P4] InP/GaSb core-shell nanowires: A novel hole-based platform with strong spin-orbit coupling for full-shell hybrid devices
A. Vezzosi, **C. Payá**, P. Wójcik, A. Bertoni, G. Goldoni, E. Prada, and S. D. Escribano
SciPost Physics, 18, 069 (2025), arXiv:2405.07651
Citations: 4

[P3] Absence of Majorana oscillations in finite-length full-shell hybrid nanowires
C. Payá, P. San-Jose, C. J. S. Martínez, R. Aguado, and E. Prada
Physical Review B, 110, 115417 (2024), arXiv:2407.06330
Citations: 8

[P2] Phenomenology of Majorana zero modes in full-shell hybrid nanowires
C. Payá, S. D. Escribano, A. Vezzosi, F. Peñaranda, R. Aguado, P. San-Jose, and E. Prada
Physical Review B, 109, 115428 (2024), arXiv:2312.11613
Citations: 21

[P1] Theory of Caroli–de Gennes–Matricon analogs in full-shell hybrid nanowires
P. San-Jose, **C. Payá**, C. M. Marcus, S. Vaitieknas, and E. Prada
Physical Review B, 107, 155423 (2023), arXiv:2207.07606
Citations: 25

Conferences

Tutorials

[C17] **Feb. 2026**. International Workshop on Superconductor Semiconductor Hybrids
Institut Néel, CNRS, Villard-de-Lans, France
Tutorial title: *Realistic Modeling of 1D Hybrid Superconductors*

Contributed Talks

[C16] **July 2025**. QTYR25
PhD and Young Scientsits in Quantum Technologies Network (PYSQT), Madrid, Spain
Contributed title: *Fluxoid valve effect in full-shell nanowire Josephson junctions*

[C15] **Mar. 2025**. APS Global Physics Summit 2025
American Physical Society, Anaheim, CA, US
Contributed title: *Josephson effect and critical currents in topological full-shell hybrid nanowires*

[C14] **July 2024**. QTYR24
PhD and Young Scientists in Quantum Technologies Network (PYSQT), Madrid, Spain
Contributed title: *Phenomenology of Majorana zero modes in full-shell hybrid nanowires*

[C13] **May 2024**. Condensed Matter PhD Program Anual Meeting
Facultad de Ciencias, Universidad Autónoma de Madrid (UAM), Madrid, Spain
Contributed title: *Phenomenology of Majorana zero modes in full-shell hybrid nanowires*

Poster contributions

[C12] **Feb. 2026**. International Workshop on Superconductor Semiconductor Hybrids
Institut Néel, CNRS, Villard-de-Lans, France
Poster title: *Non-Hermitian Skin Effect and Electronic Nonlocal Transport*

[C11] **July 2025**. Quantum Designer 2025
Donostia International Physics Center (DIPC), San Sebastián, Spain
Poster title: *Josephson effect in topological full-shell hybrid nanowires*

[C10] **June 2025**. Workshop on Superconductor-Semiconductor Hybrids
Niels Bohr Institute, University of Copenhagen, Copenhagen, Denmark
Poster title: *Josephson effect in topological full-shell hybrid nanowires*

[C9] **Apr. 2025**. GRC on Hybrid Superconductor-Semiconductor Devices
Gordon Research Conferences, Les Diablerets, Switzerland
Poster title: *Josephson effect in topological full-shell hybrid nanowires*

[C8] **July 2024**. Quantum Designer 2024
Donostia International Physics Center (DIPC), San Sebastián, Spain
Poster title: *Phenomenology of Majorana zero modes in full-shell hybrid nanowires*

[C7] **May 2024**. Quantum matter for Quantum Technologies Workshop
SPICE, Mainz, Germany
Poster title: *Phenomenology of Majorana zero modes in full-shell hybrid nanowires*

[C6] **Apr. 2024**. European School on Superconductivity and Magnetism in Quantum Materials
SuperQmap COST action, Gandía, Spain
Poster title: *Phenomenology of Majorana zero modes in full-shell hybrid nanowires*

[C5] **Sept. 2023**. Emergence of Quantum Phases in Novel Materials
Instituto de Ciencia de Materiales de Madrid (ICMM), CSIC, Madrid, Spain
Poster title: *Majorana zero modes in full-shell hybrid nanowires*

[C4] **June 2023**. Bound States in Superconducting Nanodevices
TopSquad and AndQC collaborations, Budapest, Hungary
Poster title: *Theory of Caroli-de Gennes-Matricon analogs in full-shell hybrid nanowires*

[C3] **May 2023**. QuantumMatter 2023
Phantoms Foundation, Madrid, Spain
Poster title: *Theory of Caroli-de Gennes-Matricon analogs in full-shell hybrid nanowires*

[C2] **May 2023**. YouMat2023
Instituto de Ciencia de Materiales de Madrid (ICMM), CSIC, Madrid, Spain
Poster title: *Theory of Caroli-de Gennes-Matricon analogs in full-shell hybrid nanowires*

[C1] **July 2022**. Frontiers in Condensed Matter Physics
Niels Bohr Institute, University of Copenhagen, Copenhagen, Denmark
Poster title: *Theory of Caroli-de Gennes-Matricon analogs in full-shell hybrid nanowires*

Teaching

Supervised students

Sep 2024 - June 2025 César Robles - Bachelor's Thesis
Main supervisor: Elsa Prada

Title: *Quasi-Majoranas in inhomogeneous full-shell hybrid nanowires*

Funding

Student grants

[F3] *PhD Fellowship* at ICMM (CSIC) (AEI, PRE2022-101362 for the period 2023-2027)

Participation in funded projects

[F2] *Correlations, Superconductivity and Topology in Quantum Materials and Technologies* at ICMM (CSIC) (AEI, PID2021-125343NB-I00 for the period 2022-2025). Principal investigators: Ramón Aguado and Elena Bascones

[F1] *Topology and Correlations in Quantum Materials and Solid State Quantum Technologies* at ICMM (CSIC) (AEI, PGC2018-097018-B-I00 for the period 2021-2022). Principal investigators: María José Calderón and Ramón Aguado

Outreach

March 2023 - Present ICMM Superconductivity Outreach Team
Lecturer and workshopper

Regular activities: monthly talks and demonstrations for high-school students.

Events

[O9] **Mar. 2026**. Feria Madrid es Ciencia 2026 (Madrid Science Fair)
Comunidad de Madrid, Madrid, Spain
Role: *workshopper*

[O8] **Sept. 2025**. European Researchers' Night 2025
CSIC, Madrid, Spain
Role: *workshopper*

[O7] **Mar. 2025**. Feria Madrid es Ciencia 2025 (Madrid Science Fair)
Comunidad de Madrid, Madrid, Spain
Role: *workshopper*

[O6] **Sept. 2024**. European Researchers' Night 2024
CSIC, Madrid, Spain
Role: *workshopper*

[O5] **Mar. 2024**. Feria Madrid es Ciencia 2024 (Madrid Science Fair)
Comunidad de Madrid, Madrid, Spain
Role: *workshopper*

[O4] **Nov. 2023**. Semana de la Ciencia (Science Week)
CSIC, Madrid, Spain
Role: *workshopper*

[O3] **Sept. 2023**. European Researchers' Night 2023
Instituto de Ciencia de Materiales de Madrid (ICMM), CSIC, Madrid, Spain
Role: *workshopper*

[O2] **June 2023**. Ciencia en la Calle
Casa de la Ciencia, Ciudad Real, Spain
Role: *workshopper*

[O1] **Sept. 2022**. European Researchers' Nigth 2022
Instituto de Ciencia de Materiales de Madrid (ICMM), CSIC, Madrid, Spain
Role: *logistics*

Awards

Awards and fellowships

[A4] *Max Mazín Award*. Max Mazín Foundation and CEIM Foundation, 2021. Received during the period 2018-2021. Awarded each year of my undergraduate studies, 5th to 8th editions

[A3] *GEFES Research Award for Students*. Condensed Matter Physics Division, Spanish Royal Society of Physics (GEFES-RSEF), 2020. For the work entitled *Josephson Junctions in Full-shell Majorana Nanowires*.

[A2] *Excellence Fellowship*. Comunidad de Madrid, 2018

[A1] *Premio Extraordinario de Bachillerato*. Comunidad de Madrid, 2017. Top 10 academic records of the region

Skills

Languages

Spanish	Native
English	C1 Advanced (Cambridge Certificate)
French	C1 Advanced (DALF)

Programming

OS	MacOs, Linux (Debian and Ubuntu), Windows
Programming Languages	Python (advanced), Julia (advanced), C/C++ (basic), MySQL (basic)
Web Development	HTML5, CSS, JavaScript (basic)
Scientific computing	Mathematica (advanced), Quantica (Julia package, advanced), Numpy, Scipy
Miscellaneous	Makie (Julia package, advanced), matplotlib, Git, LaTeX, Office

References available on request.